



Predictive Maintenance Analytics for Industrial Equipment

Bruno Mascarenhas da Silva

Chandan Rana

Faizanfarid Malek

Roshan Tuladhar

Submission Date: 25 January 2026

Professor: Hany Osman

Introduction

Our capstone project uses real machine sensor data to develop a predictive maintenance analytics system for industrial equipment. The primary objective is to demonstrate how advanced data analytics and machine learning techniques can identify potential machine failures before they occur.

Unexpected equipment failures remain a significant challenge in modern manufacturing environments, leading to production delays, increased operational costs, and safety risks. Traditional maintenance strategies often rely on reactive repairs or fixed maintenance schedules that do not reflect the true condition of machines.

This research simulates a real-world predictive maintenance framework by analyzing operational sensor data and developing predictive models that estimate failure risk. The project further incorporates ensemble learning, deep learning benchmarking, and model explainability techniques to support transparent and data-driven maintenance decisions. The work aligns strongly with Industry 4.0 principles, where intelligent, connected, and data-driven systems are transforming industrial operations.

Problem Identification

In many industrial environments, maintenance decisions are still based on reactive responses or fixed service schedules rather than the actual operating condition of machines. Equipment is frequently serviced too early, too late, or only after a failure has occurred. This approach results in unnecessary maintenance costs, unexpected downtime, safety concerns, and inefficient resource allocation.

A key challenge for organizations is the lack of reliable predictive tools that can detect early warning signals of machine failure under class-imbalanced conditions. Because actual failures are relatively rare events, traditional models often struggle to accurately identify high-risk machines, leading to missed failures that can be operationally costly.

Furthermore, many predictive systems operate as black boxes, providing risk scores without explaining the underlying drivers of failure. This lack of interpretability reduces trust among maintenance engineers and limits real-world adoption.

Aim of Investigation

The aim of this investigation is to develop and evaluate predictive maintenance models that estimate the probability of machine failure using operational and sensor-based data from the AI4I 2020 dataset.

Specifically, this study aims to:

- identify key sensor variables that influence machine failure risk
- comparing baseline, ensemble, and deep learning models
- optimize models under a safety-driven recall constraint
- provide explainable predictions using SHAP
- develop a production-ready predictive maintenance dashboard

The study emphasizes precision–recall performance due to the imbalanced nature of machine failures and prioritizes minimizing missed failures while maintaining acceptable false alarm rates.

RESEARCH QUESTION

Which machine learning approach provides the best precision–recall trade-off for predicting machine failure under class imbalance, and which sensor variables most strongly drive failure risk in the AI4I 2020 dataset?

Temperature and Load Interaction

H1: There is a significant probability of failure due to heat dissipation (HDF) when the difference between process temperature and air temperature falls below a critical threshold under high load conditions.

H0: The difference between process and air temperature has no significant effect on the probability of heat dissipation failure.

Sensor Association

H1: Variables related to temperature, torque, rotational speed, and tool wear are significantly associated with machine failure occurrence.

H0: These operational variables have no significant association with machine failure.

Model Performance

H1: Ensemble learning models (e.g., Soft Voting) achieve superior precision–recall performance compared to logistic regression and deep learning models for predicting machine failure.

H0: Ensemble models do not demonstrate statistically significant improvement over logistic regression or deep learning models.

Proposed Data Sources

This research uses the AI4I 2020 Predictive Maintenance Dataset obtained from Kaggle. The dataset contains approximately 10,000 observations of machine operating conditions, including:

- air temperature
- process temperature
- rotational speed
- torque
- tool wear
- product type
- machine failure indicator

The dataset is well-structured, publicly available, and suitable for supervised machine learning, ensemble modeling, and predictive maintenance research.

Intended Outcome

The intended outcome of this project is the development of a reliable and interpretable predictive maintenance system capable of identifying high-risk machine conditions before failures occur.

The project will:

- produce a high-performing ensemble model optimized for recall-sensitive environments
- benchmark deep learning performance against classical methods
- provide SHAP-based explanations of failure drivers
- deliver an interactive dashboard for real-time risk monitoring

These outputs will support maintenance planners, operations managers, and industrial engineers in implementing proactive maintenance strategies, reducing unplanned downtime, and improving resource utilization.

Overall, the study demonstrates how advanced analytics and explainable machine learning can enhance decision-making in modern Industry 4.0 environments.

References

- Kaggle. (2020). *Predictive Maintenance Dataset (AI4I 2020)*. <https://www.kaggle.com>
- Mobley, R. K. (2002). *An introduction to predictive maintenance (2nd ed.)*. Butterworth-Heinemann.
- Zonta, T., da Costa, C. A., da Rosa Righi, R., de Lima, M. J., da Trindade, E. S., & Li, G. P. (2020). *Predictive maintenance in the industry 4.0: A systematic literature review*. *Computers & industrial engineering*, 150, 106889.
<https://doi.org/10.1016/j.cie.2020.106889>
- Carvalho, T. P., Soares, F. A. A. M. N., Vita, R., Francisco, R. P., Basto, J. P., & Alcalá, S. G. S. (2019). *A systematic literature review of machine learning methods applied to predictive maintenance*. *Computers & industrial engineering*, 137, 106024.
<https://doi.org/10.1016/j.cie.2019.106024>